

# **A Symmetric Trend Veto Is Two Different Objects: 5.7 Years of Barrier Outcomes on Twenty Cryptocurrency Perpetuals**

*The short side and long side of a symmetric plus or minus 3 percent trailing return filter are not the same object, and the asymmetry is not a barrier geometry artifact*

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## **Abstract**

A common risk control on a directional trading book is a trailing return veto: do not open a new position against an extended move, no new short once the market has run up and no new long once it has sold off. This paper measures both sides of one such filter, a symmetric plus or minus 3 percent trailing 24 hour index veto, on 1,698,988 hourly take profit and stop loss brackets across twenty cryptocurrency perpetual futures from January 2021 to September 2026. The study is unconditional: every hour, every coin, both sides, no signal selection. The two halves of the filter turn out to be different objects. Take the short side first. It blocks a new short once the market has already risen by 3 percent or more, and the shorts it blocks win 1.74 percentage points less often than the shorts it still allows. That gap holds its sign in 5 of 6 calendar years, at a one sided day clustered bootstrap P of 0.974. The long side veto blocks a new long once the market has fallen by 3 percent or more, and there the blocked trades are no worse than the ones allowed, plus 0.28 percentage points, correct sign in only 2 of 6 years, P 0.386, at the live 24 hour window. The asymmetry is not an artifact of the barrier distances: the primary geometry

places the stop nearer than the target, which sets the unconditional win rate at 47.5 percent, not 50, yet re-resolving at three barrier symmetric geometries where the base rate returns to 50 percent leaves the short contrast negative. The sign is also stable. It stays negative at every threshold from 2 to 5 percent, at every trailing window from 12 to 168 hours, and when any single one of the twenty coins is dropped, and the two sides differ by 2.02 percentage points at one sided probability 0.958. None of this is a stack of independent tests. The cuts re-slice one sample, so each is marginal on its own, the six threshold family carries a familywise probability of 0.124, and once the overlapping entries are removed the short contrast falls to minus 0.27 percentage points and stops clearing the bar. What holds it up is the sign, negative across all of these cuts and correct in five of the six calendar years, a binomial probability of 0.109, not the strength of any one number. The long side loses in every trailing return state, 44 to 47 percent, so removing its veto buys candidate frequency, not edge. The claim is unconditional and about direction selection, not profitability; no state at this geometry is profitable after fees. The result explains a live asymmetry documented in the companion papers: the short side veto earns its place as exposure control, and the long side mirror was carried on faith.

## **1. Introduction**

A leveraged directional book that trades both sides of a market often carries a trailing return veto. The rule is simple: if the market has already moved a long way in some direction, do not open a fresh position against that move. On the book that motivates this paper the rule was symmetric. It blocked a new short whenever a trailing 24 hour index return sat at or above plus 3 percent, and it blocked a new long whenever that return sat at or below minus 3 percent. The same threshold, the same window, both sides. This paper asks whether the two sides deserve the same treatment, and finds that they do not.

Over 5.7 years and 1,698,988 hourly brackets, the short side of the veto blocks a genuinely worse than average group of trades. The shorts it blocks win 47.97 percent of the time; the shorts it allows win 49.71 percent; the gap of minus 1.74 percentage points holds its sign in 5 of the 6 calendar years the sample spans, and a day clustered bootstrap on the gap itself puts it below zero with one sided probability 0.974. The long side blocks a group that is, if anything, marginally better than average: blocked longs win 45.90 percent, allowed longs win 45.62 percent, a gap of plus 0.28 percentage points with the correct sign in only 2 of 6 years and one sided probability

0.386. One rule, one threshold, two different objects. On its own the short side gap is a marginal thing. Its two sided interval touches zero, it does not survive a familywise correction across the threshold family at rotation null probability 0.124, and it fades to minus 0.27 percentage points once the overlapping entries are removed. What makes the case is not that one number but the way its sign holds up across cut after cut and year after year, which Sections 5 and 6 develop.

The reader should not take any of this as evidence that either state is profitable. At the geometry studied, a take profit at plus 2.1 percent gross against a stop at minus 1.9 percent gross with a 0.10 percent round trip fee, almost nothing is profitable. The unconditional win rate is 47.5 percent, and the only trailing states with a positive net outcome anywhere in 5.7 years are shorts taken into mild weakness, which the live veto already allows. The finding is about which side of a filter carries information, not about making money. A veto that blocks worse than average trades is doing useful work even when the trades it blocks would have lost less than the ones it keeps; a veto that blocks a random subset is spending opportunity for nothing measurable.

This is the sixth in a series of working papers that place a multi month cryptocurrency trading research program on the record, mostly as negative and boundary results (Dashyan, 2026a, 2026b, 2026c, 2026d, 2026e). The companion on risk control (Dashyan, 2026c) argues that the durable edge in that program was defensive rather than predictive; this paper sharpens one of those defenses by separating the part of it that carries information from the part that does not. The companion on the backtest to live gap (Dashyan, 2026d) reports that the deployed book using this veto lost money in live trading; Section 8 returns to that, because the short side veto's value there was exposure control during a correlated selloff, not trade level prediction, and the distinction is exactly the one this paper measures.

## **2. Background and related work**

A trailing return veto is a close relative of time series momentum, the finding that an asset's own past return predicts its next return over horizons of a month to a year (Moskowitz, Ooi and Pedersen, 2012; Hurst, Ooi and Pedersen, 2017). Time series momentum says follow the move; a symmetric veto says do not fade it. The two are the same instinct at different horizons, and the horizon here is short, a trailing 24 hours, where the evidence for return continuation is thin and the evidence for reversal is at least as strong.

The more relevant literature is on the asymmetry of trend and momentum exposure. Momentum strategies crash after sharp reversals, and the crashes are concentrated in the recovery from market declines, which makes the payoff to trend exposure conditional on the sign and state of the market (Daniel and Moskowitz, 2016). Ilmanen (2011, chapters 13 and 20) frames carry and trend as strategies with built in crash asymmetry, where the same nominal rule behaves differently on the up and down side of the distribution. The result in this paper is a small, concrete instance of that asymmetry: a filter that treats the two tails identically is measured to be useful on one tail and inert on the other.

On method, the paper follows the verification standards developed across this series and the wider literature on backtest reliability. A win rate defined by a barrier race has a base rate set by the barrier distances, not by 50 percent, so every comparison is made within a side and against that side's own base rate. Multiple hypothesis concerns are handled by keeping the author chosen tests to the two vetoes and reporting the descriptive ladder as descriptive (Harvey, Liu and Zhu, 2016; White, 2000; Bailey, Borwein, Lopez de Prado and Zhu, 2014). Because hourly entries overlap in time, all inference uses a day clustered bootstrap over whole trading days instead of a binomial interval on overlapping rows (Politis and Romano, 1994; Lo and MacKinlay, 1990). The deflated Sharpe ratio and the backtesting protocol literature (Bailey and Lopez de Prado, 2014; Arnott, Harvey and Markowitz, 2019) inform the reporting discipline, though this paper reports no Sharpe ratio because it sequences no capital. The instrument, cryptocurrency perpetual futures, is the one studied by Makarov and Schoar (2020) for arbitrage and by the companion papers for direction.

None of the individual pieces is new. The contribution is the measurement: a large, unconditional, multi year barrier study that separates the two sides of a common risk filter and shows that one side carries information and the other does not, with the asymmetry surviving a barrier geometry control.

### **3. Data and method**

#### **3.1 Instrument and universe**

The instrument is the perpetual futures contract, cash settled with a funding mechanism, on twenty cryptocurrencies that a live trading program traded during 2026. The universe is fixed by

that program and is listed in full in the analysis scripts. It spans from large capitalization names to smaller alternatives. Each coin enters the study at its Binance listing date, so the panel grows from 13 coins present in early 2021 to all 20 by 2025. This is a real universe with real listing dates, not a survivorship filtered set chosen after the fact.

### **3.2 The bracket, the geometry and the account model**

The study does not simulate a sequenced book. It resolves, at every hour and for every coin and both directions, a single take profit and stop loss bracket, and records which barrier was hit first. This is the outcome oracle: for any hour, coin and side, what a trade opened then would have earned. There is no position sizing, no capital, no compounding and no concurrency, so the paper reports no return on investment, no drawdown and no Sharpe ratio. It reports win rates and per trade net outcomes only, which is the right unit for a question about direction selection.

The geometry is the live one. The take profit sits at plus 2.1 percent gross and the stop at minus 1.9 percent gross; after a 0.10 percent round trip fee the net outcomes are plus 2.00 percent on a win and minus 2.00 percent on a loss. There are no time exits: a position runs until it touches a barrier or the 30 day cap. Resolution uses 15 minute high and low bars and is pessimistic, so a single bar whose range spans both barriers is scored as the stop. Entry is the close of the 15 minute bar ending on the hour, and the barrier walk begins on the next bar, so no part of the entry bar resolves the trade it opened. An entry that never touches a barrier within 30 days is an open end and is dropped from every statistic; counting it as a scratch win would break the breakeven arithmetic. Of 1,700,000 candidate brackets, 99.979 percent resolve and 362 open ends are dropped, leaving 1,698,988 resolved brackets over 2,070 trading days from 2 January 2021 to 2 September 2026.

### **3.3 The base rate is 47.5 percent, not 50**

The barrier distances are not equal. The stop sits 1.9 percent away and the target 2.1 percent away, so a price path with no drift touches the nearer stop more often. The unconditional win rate of the geometry is 1.9 divided by the sum of 2.1 and 1.9, which is 47.5 percent, and the pooled measurement over all 1.7 million brackets is 47.5 percent, confirming the arithmetic. The gross geometry is fair, in the sense that a win pays as much as a loss costs after fees, so the profit and loss breakeven win rate is 50.00 percent; the 2.5 point gap between the 47.5 percent base rate

and the 50 percent breakeven is precisely the 0.10 percent fee. A cell at 47.5 percent is therefore informationless, not merely unprofitable, and a cell must never be read against 50 percent. Every comparison in this paper is made within a side, against that side's own base rate. Split by side over this era, the base rate is 45.67 percent for longs and 49.37 percent for shorts: the alternative coin universe drifted down across 2021 to 2026, so shorting was structurally the better side, which is itself a reason to compare only like with like.

### **3.4 The index and the trailing return**

The veto reads a market wide index and its trailing 24 hour return. The live book computed this index by normalizing each coin to its first price in a rolling window and averaging the resulting levels, which is a sound equal weight index over a fixed 109 day window with a constant coin set. Over 5.7 years it is not sound. Coins list at different dates, so a new coin enters the average at a level of one against incumbents at two or three and steps the index down on its listing. The normalization weight becomes each coin's cumulative appreciation since 2021, which turns the nominal equal weight index into a momentum weighted one. This study therefore uses a chained equal weight index: at each 15 minute step it takes the cross sectional mean of the simple returns of the coins present at both ends of the step and compounds it. That has no window start dependence and no listing discontinuity, and for a fixed coin set over 24 hours it is the same quantity the live formula approximates. Section 6.3 shows that the two indices agree on the veto decision 96 to 99 percent of the time on the overlap where both can be computed, and that the residual disagreement comes from the price data, not the index formula.

### **3.5 Statistics and reproducibility**

Hourly entries overlap: the median hold is a few hours, so consecutive entries share a price path and a binomial confidence interval on 1.7 million overlapping rows would be fiction. All inference uses a day clustered bootstrap that resamples whole trading days, so the shared market path inside a day is preserved. Confidence intervals on a single cell use 10,000 day resamples; the vetoed versus allowed contrast uses 20,000. Day clustering does not close the overlap entirely. With no time exits a position can run for days, and about a third of brackets resolve on a later calendar day than they open, the longest hold being 13 days. Adjacent day blocks are therefore not fully independent, so the intervals should be read as, if anything, a little too tight. A

de overlapped variant pushes on this directly, letting a coin re-enter only after its prior trade resolves and cutting the sample to 329,514 brackets. Both cells keep their direction, but the short contrast falls to minus 0.27 percentage points at one sided probability 0.633 and stops clearing the bar. Section 6.6 gives the full picture; it is the single result that most tempers the short side. The scripts that build the index, resolve the brackets, and compute the cells, contrasts and diagnostics are deterministic; the resolved bracket table regenerated from the raw bars this session is identical to the stored one on every field, and the reported cells match to the decimal.

#### 4. The shape of the two sides

Before the veto itself, consider the whole trailing return ladder, because the shape of it is the finding in miniature. Figure 1 plots the realized win rate of each side across twelve trailing return buckets, from the market down more than 5 percent on the left to up more than 5 percent on the right, with each side's own base rate drawn as a reference line and the two vetoed regions shaded.

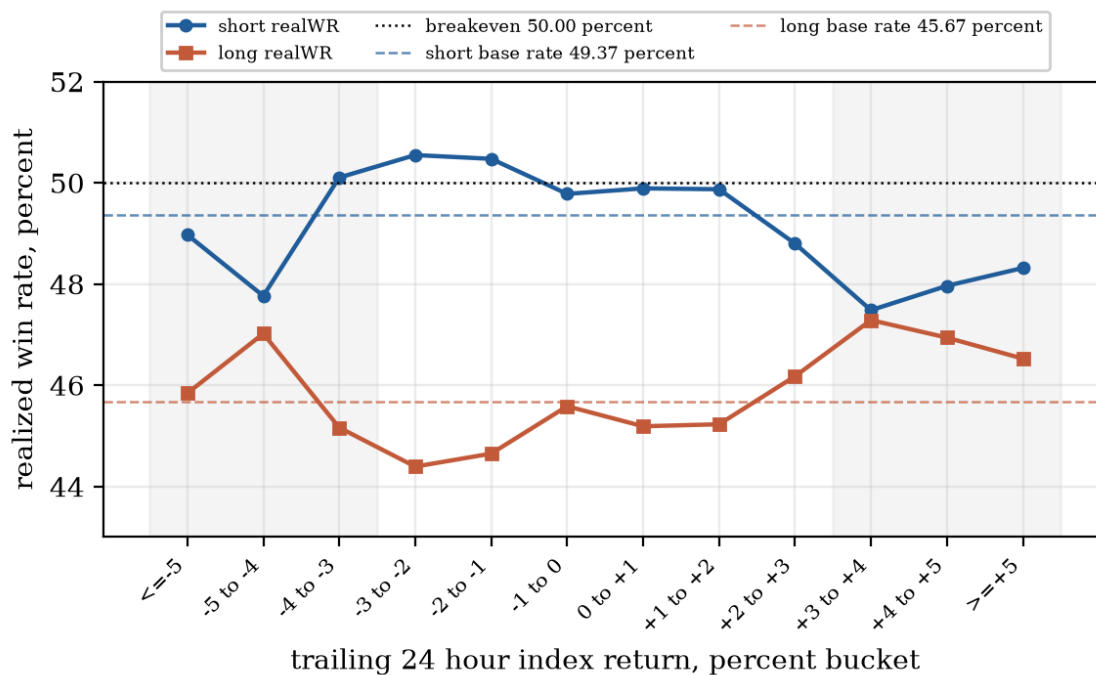


Figure 1. Realized win rate by trailing 24 hour index return, for long entries and short entries separately, over 1,698,988 brackets on 20 coins, January 2021 to September 2026. Each side is compared to its own base rate, drawn as a dashed reference line (long 45.67 percent, short 49.37 percent); the dotted line is the 50.00 percent profit and loss breakeven. The shaded bands are the vetoed states: the long veto blocks the left band (trailing return at or below minus 3 percent), the short veto blocks the right band (at or above plus 3 percent).

The shape carries the paper in two ways. First, the long line sits below its own base rate almost everywhere and never approaches breakeven, ranging from 44.4 to 47.3 percent across every bucket. Long entries lose in this universe at this geometry unconditionally; the trailing state does not rescue them and, as the next section shows, does not usefully select among them. Second, the short line is highest exactly where the market has fallen a little, from minus 4 to minus 1 percent, where it crosses breakeven, and it falls away as the market rallies. The only trailing states anywhere in 5.7 years with a positive net short outcome are those mild weakness buckets, at plus 0.004 to plus 0.022 percent per trade, and the live veto already allows them. Shorting into a rally of plus 3 percent and up is the worst short state there is, which is exactly the state the short veto blocks. The long veto, by contrast, blocks the left shaded band, where the long win rate is neither notably better nor notably worse than its flat and unremarkable level everywhere else.

This section is descriptive. It orders no cells by a selection rule and claims no significance for any single bucket; the ladder is shown so that the two vetoes can be read against the shape they sit inside. The formal tests are in Section 5.

## 5. The two vetoes and their contrast

The veto's premise is not that trades in the vetoed state lose money. Almost every state loses money at this geometry, because of the fee, and that is not a signal. The premise is that the state the veto blocks is worse for that side than the states it allows. So the decision grade number is the difference between the realized win rate of the vetoed trades and the realized win rate of the allowed trades, on the same side, with a day clustered test on the difference itself. A negative difference means the veto blocks worse than average trades and earns its place. A difference near zero means it blocks a random subset and is exposure control at best, not prediction. A positive difference means it blocks better than average trades and costs money.

Table 1 gives the two vetoed cells with their day clustered confidence intervals, the cells the gate still allows on each side, and the deep cells at plus or minus 5 percent where index measurement noise is negligible. Table 2 gives the contrast for each side.

| Cell                                                | n       | realWR | 95 percent CI  | net per trade |
|-----------------------------------------------------|---------|--------|----------------|---------------|
| Long, blocked, trailing at or below minus 3 percent | 146,458 | 45.90% | [44.16, 47.63] | -0.1641%      |

|                                                      |         |        |                |          |
|------------------------------------------------------|---------|--------|----------------|----------|
| Short, blocked, trailing at or above plus 3 percent  | 167,663 | 47.97% | [46.43, 49.56] | -0.0813% |
| Short, allowed, trailing at or below minus 3 percent | 146,458 | 49.05% | [47.33, 50.79] | -0.0380% |
| Long, allowed, trailing at or above plus 3 percent   | 167,664 | 46.86% | [45.31, 48.40] | -0.1254% |
| Long, deep, trailing at or below minus 5 percent     | 66,622  | 45.84% | [43.60, 48.20] | -0.1664% |
| Short, deep, trailing at or above plus 5 percent     | 76,267  | 48.32% | [46.36, 50.45] | -0.0673% |

*Table 1. The vetoed cells, the allowed cells and the deep cells. realWR is the realized win rate; the breakeven is 50.00 percent and the by side base rates are 45.67 percent for longs and 49.37 percent for shorts. Confidence intervals are day clustered over whole trading days, 10,000 resamples. Window January 2021 to September 2026, 1,698,988 brackets.*

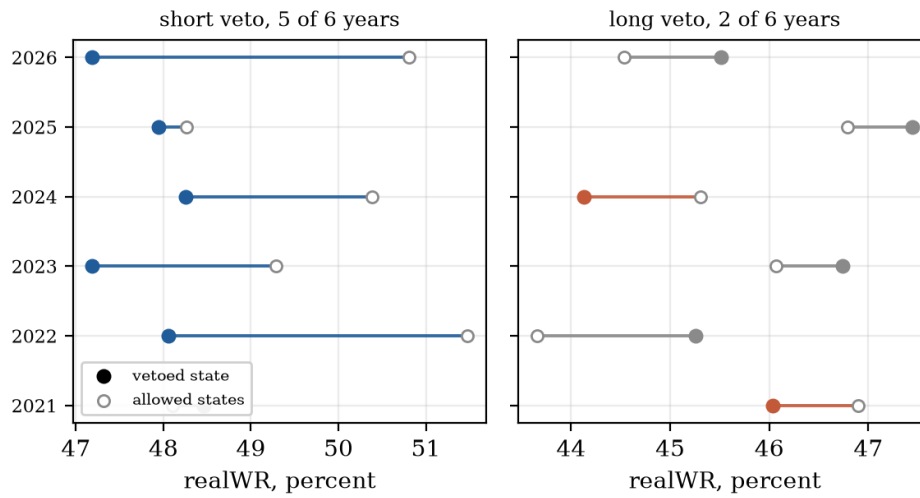
The two contrasts are the paper's central result. The short veto blocks shorts that win 1.74 percentage points less often than the shorts it allows, and the day clustered bootstrap on that gap puts it below zero with one sided probability 0.974. The program's kill convention is a one sided probability of 0.95, so the short veto clears it, though its two sided interval, from minus 3.52 to plus 0.01 percentage points, just touches zero, so it is best described as supported but marginal, not proven. The long veto blocks longs that win 0.28 percentage points more often than the longs it allows, the wrong sign for a useful veto, with one sided probability 0.386. Its blocked trades are if anything slightly better than average: minus 0.1641 percent per trade against minus 0.1753 percent for the allowed longs.

| <b>Veto</b>                                               | <b>vetoed realWR</b> | <b>allowed realWR</b> | <b>difference</b> | <b>95 percent CI</b> | <b>one sided P below zero</b> |
|-----------------------------------------------------------|----------------------|-----------------------|-------------------|----------------------|-------------------------------|
| Short, block short at trailing at or above plus 3 percent | 47.97%               | 49.71%                | -1.74 pp          | [-3.52, +0.01]       | 0.974                         |
| Long, block long at trailing at or below minus 3 percent  | 45.90%               | 45.62%                | +0.28 pp          | [-1.64, +2.20]       | 0.386                         |

*Table 2. The veto contrast: realized win rate of the vetoed trades minus the allowed trades, same side, day clustered bootstrap on the difference, 20,000 resamples. A negative difference means the veto blocks worse than average trades. Window January 2021 to September 2026.*

A pooled contrast can be carried by one regime, so Figure 2 shows the same comparison split into the six calendar years the sample spans. For each year and side it plots the vetoed win rate and the allowed win rate as a pair of points; a filled point to the left of the open point means the veto blocked worse trades that year.

Vetoed versus allowed win rate, per year, per side (a filled dot left of the open dot means the veto blocked worse trades)



*Figure 2. Vetoed versus allowed realized win rate, per calendar year, for the short veto (left) and the long veto (right). The filled point is the vetoed state, the open point the allowed states; a filled point to the left of the open point, drawn in colour, means the veto blocked worse than average trades that year, and a grey pair means it blocked better than average. Window January 2021 to September 2026, 20 coins.*

The short veto holds its sign in 5 of the 6 years, with a mean difference of minus 1.87 percentage points; only 2021 goes the other way, and by a small margin. It works in both bull years and in the 2022 bear. The long veto holds its sign in only 2 of 6 years, with a mean of plus 0.31 percentage points, flipping between blocking better and worse trades from year to year. A control whose sign is not stable across regimes is not carrying predictive content; a control whose sign is stable is. Figure 3 makes the same point through the bootstrap distributions of the two contrasts, with the observed differences and zero marked.

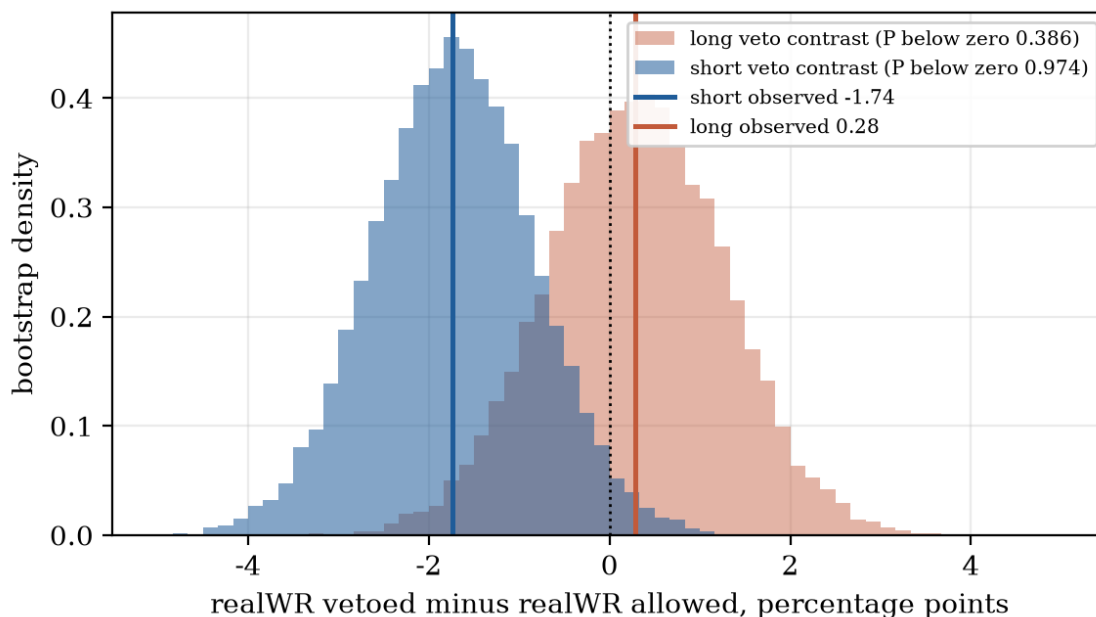


Figure 3. Day clustered bootstrap distributions of the two veto contrasts, 20,000 resamples each. The short veto distribution (blue) sits mostly below zero, observed difference minus 1.74 percentage points, one sided probability below zero 0.974. The long veto distribution (orange) straddles zero, observed plus 0.28 percentage points, probability 0.386. Window January 2021 to September 2026, 1,698,988 brackets.

What the paper does not claim here is that the short veto predicts individual trade outcomes well. A 1.74 point win rate gap is small, and the interval touches zero. The claim is narrower and, I think, more defensible: the short side of the filter blocks a genuinely worse subset in a way that is stable across six years, and the long side does not.

## 6. Robustness

### 6.1 Is the asymmetry a barrier geometry artifact

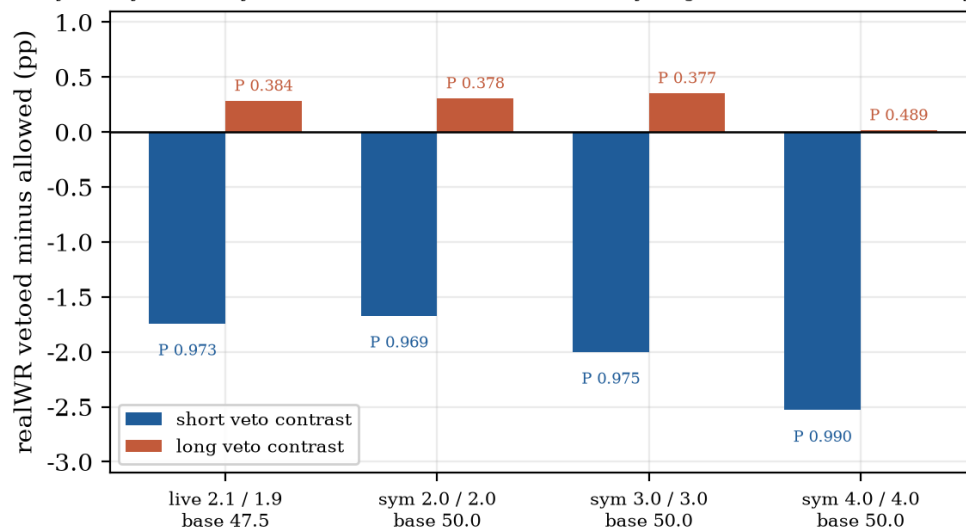
The most obvious objection is that the whole result is the 2.1 against 1.9 barrier distances. The stop is nearer than the target, the base rate is 47.5 and not 50, shorting was the better side in a falling market, and perhaps the short veto's contrast is that structure interacting with the trailing state and not anything about the trailing state itself. The test is direct: re-resolve the same 1,698,988 entries at barrier symmetric geometries, where the stop and target sit at equal distance and the base rate returns to 50 percent, and recompute both contrasts. If the asymmetry is the barrier distances, it should vanish; if it is the trailing state, it should survive.

Table 3 and Figure 4 report three symmetric geometries alongside the live one: a narrow symmetric barrier at plus or minus 2.0 percent gross, a wider one at plus or minus 3.0 percent, and a wider one still at plus or minus 4.0 percent. The live row reproduces the contrast script exactly, as a control.

| Geometry                  | base realWR | short veto difference | short P below zero | long veto difference | long P below zero |
|---------------------------|-------------|-----------------------|--------------------|----------------------|-------------------|
| Live, target 2.1 stop 1.9 | 47.52%      | -1.74 pp              | 0.973              | +0.28 pp             | 0.384             |
| Symmetric 2.0 and 2.0     | 49.96%      | -1.67 pp              | 0.969              | +0.30 pp             | 0.378             |
| Symmetric 3.0 and 3.0     | 49.99%      | -2.00 pp              | 0.975              | +0.35 pp             | 0.377             |
| Symmetric 4.0 and 4.0     | 49.99%      | -2.53 pp              | 0.990              | +0.01 pp             | 0.489             |

*Table 3. The two veto contrasts re-resolved at four geometries on the same 1,698,988 entries. At the three symmetric geometries the base rate returns to 50 percent, yet the short contrast stays negative and its one sided probability stays at or above 0.969, strengthening with barrier width, while the long contrast stays near zero. The short veto era sign test is 5 of 6 at every geometry. Window January 2021 to September 2026.*

The asymmetry survives symmetric barriers: the short contrast stays negative and its one sided P stays above 0.96



*Figure 4. The veto contrasts across four barrier geometries. As the barriers become symmetric and then wider, the base rate returns to 50 percent but the short veto contrast (blue) stays negative and deepens, one sided probability rising from 0.973 to 0.990, while the long veto contrast (orange) stays near zero. The short side asymmetry is not an artifact of the live 2.1 against 1.9 barrier distances. Same 1,698,988 entries, January 2021 to September 2026.*

The base rate does move as expected, from 47.52 percent at the live geometry to 49.96 percent and above once the barriers are equidistant, which confirms that the 47.5 figure is the barrier distance effect. But the short veto contrast does not move with it. It stays negative at every geometry, its one sided probability stays at or above 0.969, and it in fact deepens as the barriers widen, to minus 2.53 percentage points at plus or minus 4.0 percent with probability 0.990. The short veto era sign test remains 5 of 6 at every geometry. The long contrast stays small and positive throughout. The asymmetry is a property of the trailing state, not of the barrier distances.

## 6.2 The threshold is not a fitted number

The plus 3 percent threshold could be a number chosen because it happens to work. The test is to sweep it. Table 4 and Figure 5 give both veto contrasts at six thresholds from 2.0 to 5.0 percent of trailing return. The short contrast is negative at every one of them, from minus 1.60 percentage points at 2.0 percent to minus 1.15 at 5.0 percent, and the long contrast straddles zero throughout, from minus 0.29 to plus 0.65 percentage points with no stable sign.

| Threshold percent | short veto difference | short P below zero | long veto difference | long P below zero |
|-------------------|-----------------------|--------------------|----------------------|-------------------|
| 2.0               | -1.60 pp              | 0.972              | -0.29 pp             | 0.621             |
| 2.5               | -1.60 pp              | 0.967              | -0.14 pp             | 0.555             |
| 3.0 (live)        | -1.74 pp              | 0.973              | +0.28 pp             | 0.383             |
| 3.5               | -1.45 pp              | 0.940              | +0.36 pp             | 0.364             |
| 4.0               | -1.34 pp              | 0.911              | +0.65 pp             | 0.279             |
| 5.0               | -1.15 pp              | 0.848              | +0.19 pp             | 0.443             |

*Table 4. Both veto contrasts across six trailing return thresholds, day clustered bootstrap on the difference, same 1,698,988 entries. The short contrast is negative at every threshold; the long contrast has no stable sign. Window January 2021 to September 2026.*

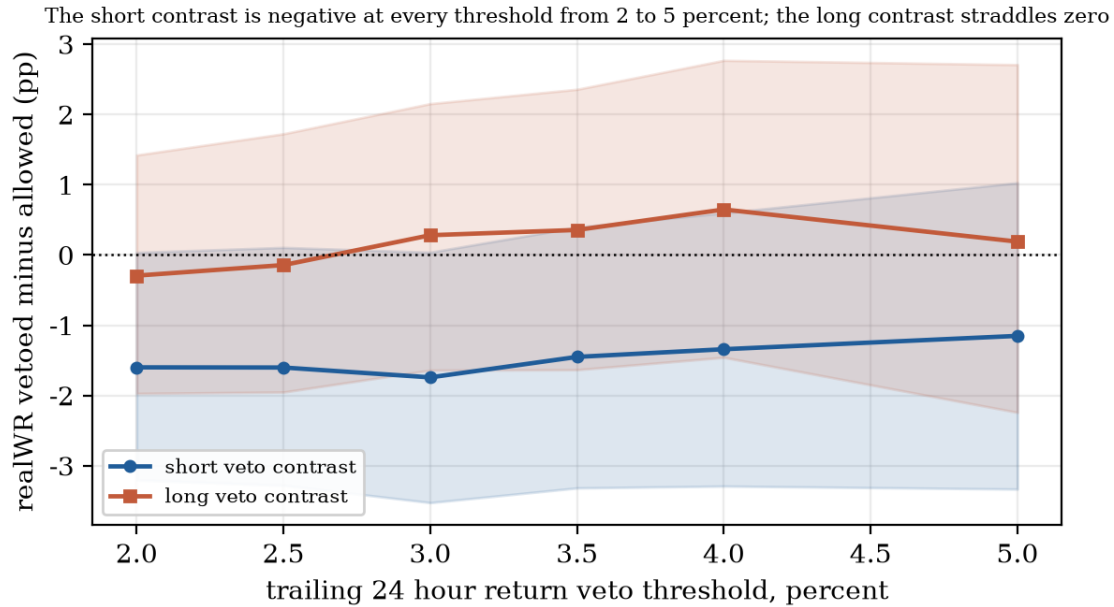


Figure 5. The veto contrast against the threshold, with day clustered confidence bands. The short contrast (blue) is below zero at every threshold from 2 to 5 percent; the long contrast (orange) straddles zero. Same 1,698,988 entries, January 2021 to September 2026.

Two honest points sit next to this. The live gate uses plus 3 percent because that is the threshold the trading system was built with, not one I chose from this sweep, so its single cut probability of 0.973 is a pre-registered test, not a fished one. But if one instead asks whether SOME threshold in this range shows a short contrast, that is a search over six correlated cuts and must be corrected. Rotating the trailing return in time, which preserves its rate and serial structure while breaking its alignment with outcomes, and taking the most negative short contrast across the six thresholds, the observed best is beaten by 12.4 percent of rotations, a familywise probability of 0.124. So the short veto does not become individually significant by searching thresholds. Its credibility is not any single cut; it is that the sign is negative at all six thresholds at once, which a search over these correlated thresholds reproduces only about 12 percent of the time, and the long side's familywise probability of 0.56 confirms the null behaves as it should. These six thresholds are nested subsets of one another, so their agreement is one correlated fact rather than six independent ones.

### 6.3 The trailing window is not special either

The live gate reads a trailing 24 hour return. Table 5 recomputes the plus or minus 3 percent contrast using 12, 24, 48 and 168 hour trailing windows. The short veto is supported at every horizon and more strongly at three of the four, with one sided probabilities of 0.999 at 12 hours, 1.000 at 48 hours and 0.997 at 168 hours against 0.975 at the live 24 hours. The short side does not depend on the window choice.

| Trailing window | short veto difference | short P below zero | long veto difference | long P below zero |
|-----------------|-----------------------|--------------------|----------------------|-------------------|
| 12 hours        | -2.91 pp              | 0.999              | -1.80 pp             | 0.968             |
| 24 hours (live) | -1.74 pp              | 0.975              | +0.28 pp             | 0.385             |
| 48 hours        | -3.15 pp              | 1.000              | -1.48 pp             | 0.942             |
| 168 hours       | -2.47 pp              | 0.997              | -1.69 pp             | 0.963             |

*Table 5. The plus or minus 3 percent veto contrast at four trailing windows, day clustered bootstrap, same brackets. The short contrast is negative at every window. The long contrast is random at the live 24 hour window and negative at the other three. Window January 2021 to September 2026.*

The long side carries a real nuance here, and it is worth being blunt about. At the live 24 hour window the long veto blocks a random subset, plus 0.28 percentage points. At 12, 48 and 168 hours it blocks a somewhat worse subset, minus 1.5 to minus 1.8 percentage points. Longs still lose in every state, so this is which losing subset the veto blocks, and no state is a profitable long. It still means the live gate applies its long veto at the one trailing horizon where that veto carries the least information. The short side has no such sensitivity. There is a consequence worth naming. The clean split between the two sides, the long side random while the short side bites, is really a property of the 24 hour horizon. The short side holds up at every window tested. The long side reads as random only at 24 hours; stretch or shorten the window and it too begins to block a worse subset, so the two sides start to look more alike. The asymmetry this paper reports, then, is the one at the horizon the live gate actually runs on.

### 6.4 No single coin carries the result

A panel of twenty correlated coins can hide a result that one coin drives. Figure 6 recomputes the plus or minus 3 percent contrast twenty times, dropping each coin in turn. The short contrast stays negative for all twenty removals, from minus 1.87 to minus 1.62 percentage points; the least supportive drop is removing FIL, and the short contrast is still minus 1.62. The long

contrast stays between plus 0.18 and plus 0.46 for all twenty. No coin carries the short asymmetry, and no coin makes the long veto look predictive.

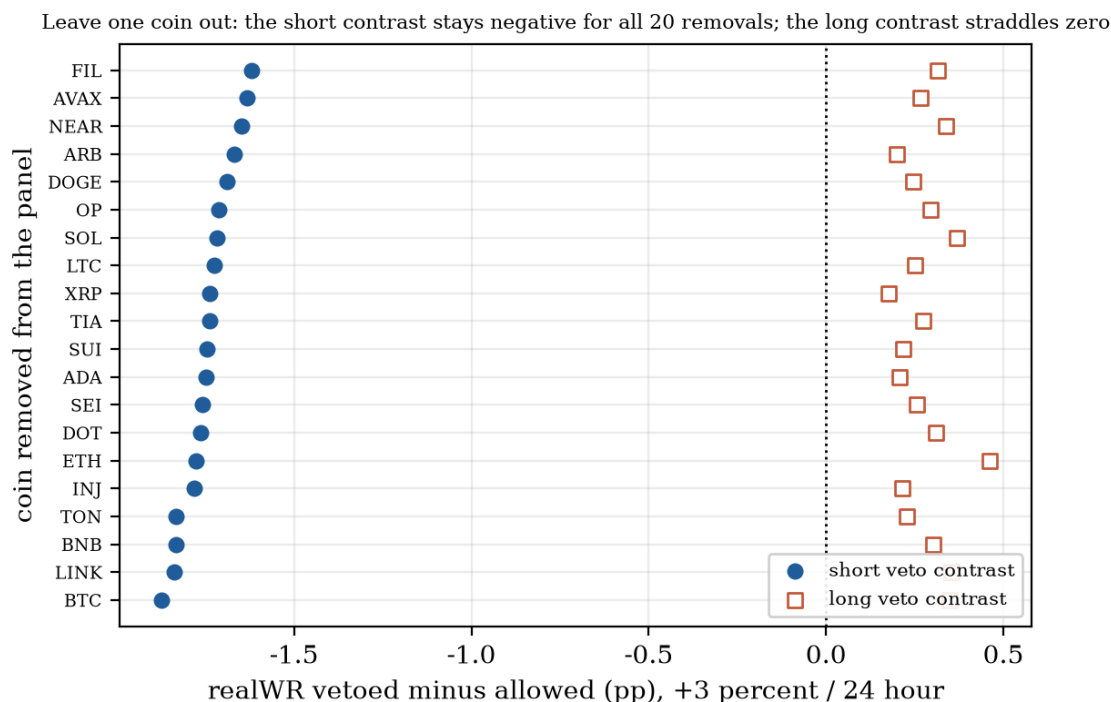


Figure 6. Leave one coin out. The plus or minus 3 percent, 24 hour veto contrast recomputed with each coin removed from the panel. Every short removal (blue) is negative; every long removal (orange) sits just above zero. Same brackets, January 2021 to September 2026.

## 6.5 The two sides are statistically distinct

The paper's central claim is that the two sides are different objects. That can be tested directly by the difference between the two contrasts, resampling whole days once and computing both contrasts on the same days so the shared market path cancels. The short contrast is minus 1.74 percentage points, the long contrast is plus 0.28, and their difference is minus 2.02 percentage points with a day clustered interval from minus 4.35 to plus 0.26 and a one sided probability of 0.958 that the short side blocks a worse subset than the long side. Alongside the era sign test, 5 of 6 years for the short side against 2 of 6 for the long, the asymmetry is the stable object even though each side's own contrast is marginal. A linear slope of win rate on the trailing return is not significant on either side, which is expected: the short effect is concentrated in the rally tail the veto blocks, not spread as a trend across the whole range, so a straight line averages it away.

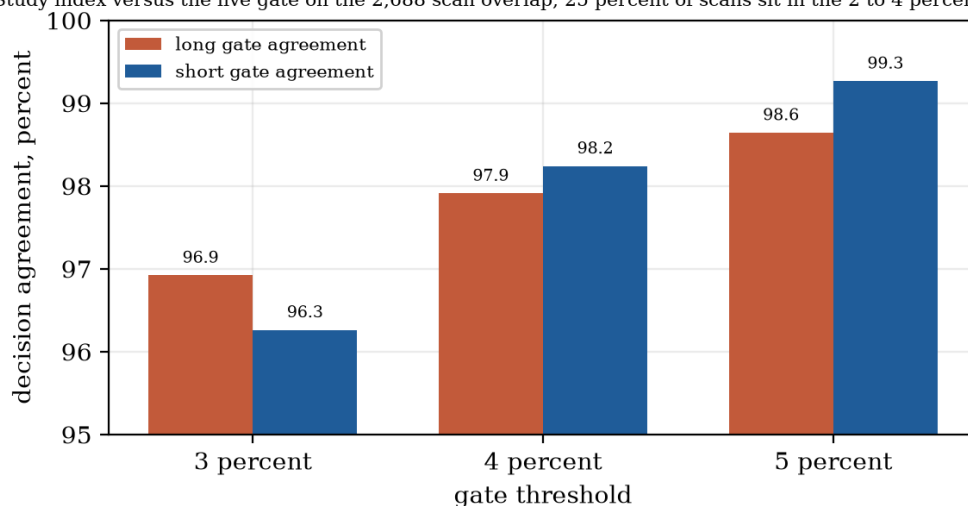
## **6.6 The deep cells, the two year window and the overlap**

Three further checks are in the underlying tables. The deep cells at plus or minus 5 percent, in Table 1, tell the same story as the 3 percent cells: long deep 45.84 percent, short deep 48.32 percent, so the finding is not a boundary artifact where the index is noisiest. The most recent two years alone, from September 2024 to September 2026, reproduce the asymmetry: the blocked longs win 47.01 percent against a 46.20 percent long base rate for that window, so again slightly better than average, while the blocked shorts win 47.55 percent against a 48.89 percent short base rate, so worse than average. Finally the de overlapped variant, which keeps only one trade per coin at a time and so strips out the within coin overlap in the hourly entries, holds the direction of both cells on 329,514 brackets: blocked longs 47.14 percent, blocked shorts 48.11 percent. The contrast on that sample is where the short result is most exposed, so I will not soften it. The short contrast falls from minus 1.74 percentage points to minus 0.27, with a day clustered interval from minus 1.74 to plus 1.22 and a one sided probability of 0.633; on this sample it no longer clears the one sided convention. The long contrast is plus 1.01 percentage points, probability 0.104. Neither sign flips, but a good part of the short contrast's apparent significance on the full sample was coming from the shared price paths of overlapping entries, which is what the day crossing in Section 3.5 already suggested. This is the main reason the paper leans on the sign of the short result across cuts, and on its exposure control role, rather than on any single contrast reaching significance.

## **6.7 Does the study index measure the live gate**

The study uses a chained equal weight index in place of the live book's window normalized one, so it must be shown that the two agree on the veto decision. On the 2,688 scan overlap where the live frame's own prices are available, the two indices' trailing 24 hour returns correlate at 0.964 with a mean absolute difference of 0.49 percentage points and a bias of plus 0.014, symmetric noise rather than a systematic offset. Figure 7 shows the decision agreement at three thresholds.

Study index versus the live gate on the 2,688 scan overlap; 25 percent of scans sit in the 2 to 4 percent fuzzy band



*Figure 7. Agreement between the study's chained index and the live gate's index on the veto decision, at the plus or minus 3, 4 and 5 percent thresholds, over the 2,688 scan overlap where both can be computed. Agreement is 96 to 99 percent and rises with the threshold as fewer decisions fall near the boundary. About 25 percent of scans sit in the 2 to 4 percent band where the plus or minus 3 percent decision is partly a coin flip.*

Agreement is 96.9 and 96.4 percent for the long and short decision at plus or minus 3 percent, rising to 98.6 and 99.3 percent at plus or minus 5 percent. The residual disagreement is not the index formula: running the live formula on the study's own price bars gives 96.9 percent, the chained index 96.8 percent, indistinguishable. The cause is the price data. The live frame's price is a scan time snapshot and the study's is the official bar close, and they differ by a per coin median of 0.09 to 0.28 percent. Because about 84 percent of the disagreements sit within one percentage point of the threshold, the effect is boundary blur, which attenuates bucket differences rather than manufacturing them, and the deep cells confirm that the attenuation did not drive the result.

This raises a finding in its own right. The live gate applies a hard plus or minus 3 percent cut to a quantity that carries about 0.76 percentage points of measurement noise, the standard deviation of the difference between the two indices, and about 25 percent of scans sit in the 2 to 4 percent band where the decision is partly a coin flip. Whatever the merits of the premise, the implementation has a fuzzy boundary, and a filter whose threshold is crossed by measurement noise a quarter of the time is closer to a soft weighting than to the hard rule it is written as.

## **7. Where the effect is absent**

The clearest way to state the long side result is that the long veto is not why long entries lose. Long entries lose everywhere. Across all twelve trailing buckets the long win rate ranges from 44.4 to 47.3 percent, never near the 50 percent breakeven and never above its own 45.67 percent base rate by enough to matter. The state the long veto blocks, the market down 3 percent or more, is not distinguishable from the states it allows: the difference is plus 0.28 percentage points, the wrong sign, with the correct sign in only 2 of 6 years.

The consequence is practical and slightly counterintuitive. Removing the long veto does not create a profitable long book, because there is no profitable long state to uncover; the best long bucket anywhere is 47.3 percent. What removing it buys is candidate frequency. The long veto fires on 17.2 percent of long candidate hours, so dropping it returns roughly one long candidate in six to the book, and the paper's finding is that those returned candidates are no worse, and no better, than the longs already taken. Whether that added frequency is worth having is a question about the book's other constraints, not about the trailing state, and this unconditional study cannot answer it. What it can say is that the long veto is a legitimate removal candidate on the evidence, the first gate change in this program that five years of data support rather than merely tolerate, and that its removal is a frequency decision, not an edge decision. It should not be replaced by a symmetric threshold at some other value, because the asymmetry, not the level, is the finding.

## **8. The effect on a live, signal selected book**

This paper is unconditional by necessity: the live book selected candidates from a set of indicator feeds that do not exist in the historical record before mid 2026, so a multi year answer can only be an all hours, all coins answer. It says whether fading or following an extended market move pays in general, not whether it pays on the small, selected set of candidates the live book actually traded. The numbers in this section are therefore cited from the program's dated internal record and the companion papers (Dashyan, 2026c, 2026d) and were not re-run for this paper; they are the application, and the 5.7 year study above is the spine.

On the live book's 109 day window, the short veto blocked 99 short candidates that had passed the book's selection, at a mean net of minus 0.5859 percent per candidate and a realized win rate

of 35.35 percent, well below the 50 percent breakeven. The long veto blocked 136 long candidates at minus 0.5996 percent and 33.82 percent. On that selected set both blocked groups were clear losers, which looks at first like support for both vetoes. But the rotation null, which rotates the veto's own time series to preserve its rate while destroying its alignment with the market, validates only the short side at the fixed threshold, at probability 0.002, and the long side mirror on its 7 blocked holdout candidates returns probability 0.2625, not distinguishable from a rate matched sham. At the day level neither side is significant, probabilities 0.42 and 0.72, which is consistent with the unconditional finding that the short veto is exposure control rather than trade level prediction: its value shows up when the market falls together, not trade by trade.

The exposure control value is real and large on the selected book, and it is the reason the short veto was kept. On 20 August 2026 the raw selection rule, before the veto, fired 88 short candidates in a single day against a typical 15, because a vertical rally pushed most of the universe into the same oversold reading at once. The veto blocked that correlated cluster. Measured on the deployed book, running the raw rule with the short veto instead of without it cut the maximum drawdown from about 71 percent to about 9 percent, roughly 62 points. It kept a book that would otherwise have lost about 44 percent from doing so. These live book figures are cited from the internal record and are not reproduced in this paper. That is what a veto worth keeping does on a leveraged book: it does not predict which single short will lose, it prevents the book from taking eighty correlated shorts into a rally on the same afternoon. The unconditional study is the mechanism behind that number. Shorting into a rally is the worst short state there is, and the veto blocks exactly it.

The honest limit of the selected book evidence is that a threshold sweep across its 109 days is a plateau, not a cliff. No single threshold cell in a grid of 66 clears the program's significance bar. And the deployed book that used this veto lost money in live trading, as the companion on the backtest to live gap documents (Dashyan, 2026d). The short veto did not make that book profitable. It made it survivable. Those are different claims, and only the second one is supported.

## **9. Discussion and limitations**

The limitations are ordered by how much they bound the claim.

First, the study is one universe and, primarily, one geometry. The twenty coins are a real traded set with real listing dates, but they are correlated crypto alternatives in a single asset class, and the down drift of that universe across 2021 to 2026 is why the short base rate sits above the long. The barrier geometry robustness in Section 6.1 addresses the sharpest version of the geometry objection, but it does not make the result multi asset. Whether the same asymmetry appears in equity index futures or foreign exchange is unknown and would be the natural next test.

Second, the study is unconditional, and the live book was conditional. The 5.7 year answer and the 109 day selected book answer agree that the long veto is not predictive and the short veto is supported, from two different data spans and two different methods, which is the strongest form the evidence takes. But the unconditional study cannot speak to the selected book's profitability, and the selected book's numbers in Section 8 are cited, not reproduced here.

Third, the effect is small and marginal in any single test. The short contrast is 1.74 percentage points with a two sided interval from minus 3.52 to plus 0.01, supported at the program's one sided convention and no more. When the plus 3 percent threshold is treated as one of a family of six and corrected by a rotation null, the best short contrast is not individually significant, familywise probability 0.124, and I would rather say so than dress it up. The value of the finding is not the size or the significance of any one cut. It is that the short contrast is negative at every threshold from 2 to 5 percent, at every trailing window from 12 to 168 hours, when any single coin is dropped, and at three barrier symmetric geometries, and that its sign is right in 5 of 6 years while the long contrast wanders through all of the same cuts. It would be easy to oversell that. These cuts are not independent draws: they re-slice one set of brackets, the thresholds nest inside one another, the windows are autocorrelated, the leave one coin out panels share nineteen of twenty coins, and the geometries re-price the same paths. Only the six calendar years are close to disjoint, and there the short sign is right in five, a one sided binomial probability of 0.109. Take the overlap out altogether and the short contrast slips to minus 0.27 percentage points and stops clearing the bar. So the argument is not that a search could never produce this sign; a search over the correlated thresholds produces it about 12 percent of the time. It is that one fished threshold would not satisfy all of these cuts at once, that the sign is right in five of six near independent years, and that a difference in differences separates the two sides at probability 0.958. That pattern is the evidence, not any single p value, and it is suggestive rather than conclusive.

Fourth, listing effects and index construction. Coins enter at their listing dates, so the early years carry fewer coins and the index is thinner then; the chained construction handles the discontinuity but cannot add history that does not exist. The choice of a chained equal weight index over the live formula is defended in Section 3.4 and validated in Section 6.3, but it is a choice, and a reader who preferred the live formula would find the live formula agrees with it on 96 to 97 percent of decisions.

Fifth, the implementation's fuzzy boundary. The 0.76 percentage points of index measurement noise and the 25 percent of scans in the 2 to 4 percent band mean the live gate is a softer rule than its hard threshold suggests. This attenuates the measured contrasts instead of inflating them, so it makes the reported effects conservative, but it also means the exact threshold is less meaningful than it looks.

This paper makes no trading recommendation. It reports which side of a filter carries information on one universe over one span, at a geometry where nothing is profitable after fees. Nothing here is advice to trade any instrument, on either side, with or without the veto.

## **10. Conclusion**

A symmetric trailing return veto turns out to be two different objects. Over 5.7 years and 1.7 million barrier outcomes, the short side of a plus or minus 3 percent trailing 24 hour veto blocks shorts that win 1.74 percentage points less often than the shorts it allows. The sign is stable in 5 of 6 years, and the contrast stays negative at every threshold from 2 to 5 percent, at every trailing window from 12 to 168 hours, dropping any single coin, and at three barrier symmetric geometries where the base rate returns to 50 percent. Every one of these cuts is marginal, and because they re-slice a single sample they are not independent tests; the effect does not survive a familywise correction across the threshold family at the 0.05 level, and it slips below the bar once the overlapping entries are removed. Its weight lies in how steadily the sign holds, across the cuts and across the six years, not in any one of them. The long side blocks a subset no different from average, plus 0.28 percentage points, correct sign in 2 of 6 years, probability 0.386. The long side loses in every trailing state, so removing its veto buys candidate frequency rather than edge; the short side veto earns its place, but as exposure control that stops a book from taking many correlated shorts into a rally, not as a predictor of individual trades. The general lesson is narrow and, I hope, transportable: a risk filter that treats two tails identically

should be measured on each tail separately, because the same nominal rule can carry information on one side and none on the other.

## **Author Contributions and Use of AI Tools**

The author designed the study, specified the universe, geometry, thresholds and units of analysis, made every modelling and inferential decision, and takes full responsibility for the contents of this paper including any errors.

Large language model assistants were used as instruments under the author's direction, in three roles. They wrote and executed the analysis code from the author's specifications, including the index construction, the bracket resolver, the day clustered bootstrap and the second geometry re-resolution. They were used adversarially, in a structured effort to break the paper's claims instead of supporting them, which produced the barrier geometry control, the de overlapped independence check and the base rate correction that this paper is built around. And they assisted in drafting prose from the author's outline and results ledger. Claude is a tool, not an author.

Two safeguards were imposed. Every number in this paper is pinned to a ledger entry naming the script that produced it, and the resolved bracket table was regenerated from the raw price bars during preparation and confirmed identical to the stored one on every field. All analysis scripts are deterministic under fixed seeds, and the reported cells and contrasts reproduce to the decimal across runs. No attempt was made to defeat text detectors; the humanising work was genuine rewriting, and this disclosure is the honest account of how the paper was produced.

## **Data and Code Availability**

The price data is 15 minute open, high, low and close bars for twenty perpetual futures contracts from a public exchange endpoint requiring no key. The universe is listed in full in the analysis scripts. Five deterministic scripts build the chained index and its validation against the live gate, resolve the 1,698,988 brackets, and compute the cells, the ladder, the era table, the two year window and the de overlapped robustness, the vetoed versus allowed contrast and era sign test, and the index diagnosis. A sixth re-resolves the same entries at the three symmetric geometries, and a seventh recomputes the overlap removed contrast and the day crossing diagnostics. A results ledger records every number with its source script and its saved run. The 109 day selected

book figures in Section 8 are cited from a separate dated internal record and were not re-run for this paper; they should be read as citations, not as results reproduced here.

## References

- Arnott, R., Harvey, C. R., and Markowitz, H. (2019). A backtesting protocol in the era of machine learning. *Journal of Financial Data Science*, 1(1), 64 to 74.
- Bailey, D. H., Borwein, J. M., Lopez de Prado, M., and Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: the effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5), 458 to 471.
- Bailey, D. H., and Lopez de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94 to 107.
- Daniel, K., and Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221 to 247.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5 to 68.
- Hurst, B., Ooi, Y. H., and Pedersen, L. H. (2017). A century of evidence on trend-following investing. *Journal of Portfolio Management*, 44(1), 15 to 29.
- Ilmanen, A. (2011). *Expected Returns: An Investor's Guide to Harvesting Market Rewards*. Wiley, Chichester, chapters 13 and 20.
- Lo, A. W., and MacKinlay, A. C. (1990). Data-snooping biases in tests of financial asset pricing models. *Review of Financial Studies*, 3(3), 431 to 467.
- Makarov, I., and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293 to 319.
- Moskowitz, T. J., Ooi, Y. H., and Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228 to 250.
- Politis, D. N., and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303 to 1313.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097 to 1126.
- Dashyan, A. (2026a). Short-horizon directional non-predictability in cryptocurrency perpetual futures: a multi-method negative result under adversarial verification. SSRN Working Paper.  
<https://doi.org/10.2139/ssrn.7306538>

- Dashyan, A. (2026b). Adversarial backtest verification: catching overfitting, beta, collinearity, and look-ahead in a live trading-research program. SSRN Working Paper. <https://doi.org/10.2139/ssrn.7338919>
- Dashyan, A. (2026c). Risk control as the durable edge: loss filtering, profit-armed ratchets, and market-neutral carry when direction fails. SSRN Working Paper. <https://doi.org/10.2139/ssrn.7345542>
- Dashyan, A. (2026d). A real edge that loses: anatomy of a backtest-to-live gap in cryptocurrency perpetual futures. SSRN Working Paper. <https://doi.org/10.2139/ssrn.7353678>
- Dashyan, A. (2026e). The tail is the only signal: flush reversion in equity indices and crypto perpetual futures. SSRN Working Paper. <https://doi.org/10.2139/ssrn.7363482>